1 Probability Foundations for Data Science – Megan Silvey

LinkedIn Learning Course

1.1 Introduction

1.1.1 Derivative rules

- $\frac{d\,x^n}{dx} = n * x^{n-1}$
- $\frac{d\,f(x)+g(x)}{dx} = f'(x)+g'(x)$
- $\frac{d\,f(x)*g(x)}{dx} = f'(x)*g(x)+f(x)*g'(x)$
- $\frac{d\,\frac{f(x)}{g(x)}}{dx} = \frac{f'(x)*g(x)-f(x)*g'(x)}{g(x)^2}$
- $\frac{d\,f(g(x))}{dx} = f'(g(x))*g'(x)$

1.1.2 Integrals

- $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$, where C is an unknown value
- $\int f(x)+g(x)\,dx = \int f(x)\,dx + \int g(x)\,dx$
- $\int a*f(x)\,dx = a*\int f(x)\,dx$

1.2 Probability Fundamentals

$$P(A) = \frac{Number\ of\ \ favourable\ outcomes}{Total\ number\ of\ \ possible\ outcomes}$$

An experiment or trial is any procedure that can be infinitely repeated, well defined set of possible outcomes; the terms experiment and trial are used interchangebly.
Sample space: the set of all possible outcomes of an experiment. Event: a subset of sample space, represents one or more outcomes of an experiment, denoted as A with probability $P(A)$.
Permutations arrangements of items in a specific order.

The number of permutations of n items, taken p at a time is: $P(n,p) = \frac{n!}{(n-p)!}$

Combinations: arrangements of items, **not** in a specific order. So, AB and BA are the same.

The number of combinations of n items, taken p at a time is: $C(n,p) = \frac{n!}{p!*(n-p)!}$

Independent events. Two events are independent if the occurrence of one does not affect the probability of the other. Formula: $P(A \cap B) = P(A)*P(B)$ the formula works if and only if the events A and B are independent.
Two events are dependent if the occurrence of one **does** affect the probability of the other.

- $P(A \cap B) = P(A)*P(B \mid A)$
- $P(A \cap B) = P(B)*P(A \mid B)$

Mutually exclusive events. $P(A \cup B) = P(A)+P(B)$

2 Biography

ID	Title	Version
[LW03]	"All of Statistics A Concise Course in Statistical Inference", Larry Wasserman, 736 pages	2003